UNISYS PROJECT

TOPIC: RESQTRACK – AI-ENABLED SMART RESCUE TRIAGE BAND AND DASHBOARD SYSTEM

Abstract

Introduction

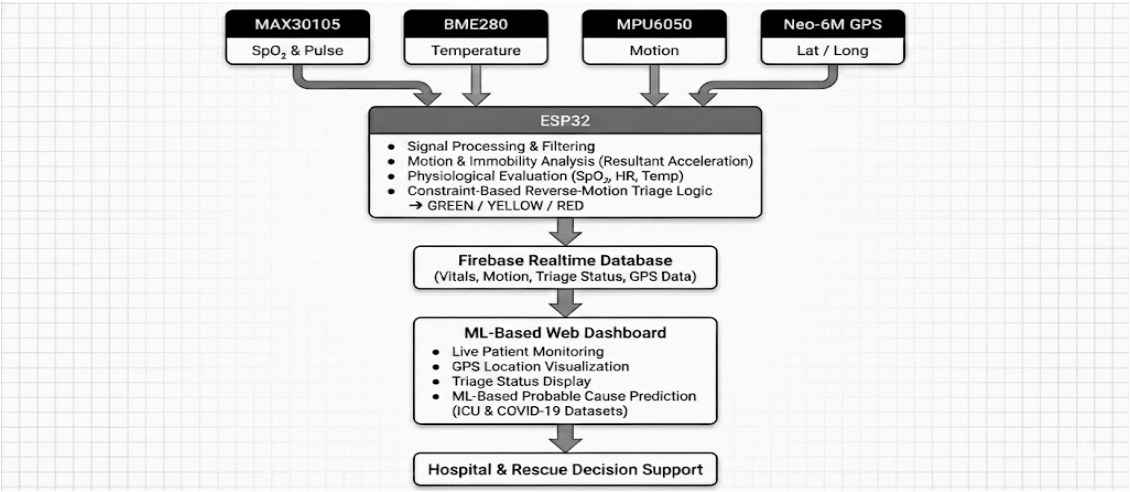
ResqTrack is a smart rescue and triage system designed as a **wearable band** for emergency situations such as accidents and disasters. It continuously monitors vital signs, patient movement, and location, helping rescue teams quickly understand a patient's condition without relying only on manual checks.

The data from the band is sent to a **shared web-based dashboard** used by hospitals and rescue teams, improving communication and coordination during rescue operations. An integrated **AI model**, trained on ICU and COVID-19 patient data, further supports decision-making by indicating patient severity and possible medical causes, enabling faster and more informed pre-hospital care.

Objectives

1. **Multi-Parameter Health Monitoring :** To design a multi-parameter patient monitoring system integrating SpO_2, heart rate, body temperature, motion status, and GPS location using MAX30105, BME280, MPU6050, and Neo-6M sensors.
2. **Constraint Based Triage Classification :** To implement a robust constraint-based triage algorithm that classifies patients into GREEN, YELLOW, and RED categories using clinically realistic thresholds.
3. **Reverse Motion Severity Detection :** To employ reverse-motion logic suitable for rescue scenarios, where prolonged immobility combined with deteriorating vitals indicates higher clinical severity.
4. **Cloud-Integrated Real-Time Monitoring :** To transmit all sensor data and triage status reliably to a Firebase Realtime Database to develop a real-time web-based dashboard with ML-assisted probable cause prediction trained on ICU and COVID-19 patient datasets.
5. **AI-Enabled Decision Support & Scalability :** To design the system with scalability, manufacturability, and energy efficiency for future wearable deployment.

Methodology



Expected Outcomes

1. **Accurate & Intelligent Triage:**Enables faster and more reliable patient prioritization during rescue operations through multi-parameter sensor fusion and inactivity-based triage logic, minimizing false positives.
2. **Real-Time Situational Awareness & Decision Support:**Provides hospitals and responders with live GPS-enabled dashboards and AI-assisted insights to support timely pre-hospital care, especially in remote or delayed-access scenarios.
3. **Scalable & Manufacturable Deployment:**Delivers a compact, energy-efficient, and area-optimized system suitable for PCB integration, large-scale manufacturing, and subscription-based deployment across emergency and healthcare services.

Mentor

Ms. Sujata Priyambada Mishra

Student Details

Gouri Prakash	1RV24EC073
Manasa M	1RV24EC114
Monika P Devigere	1RV24EC126
Siddesh Sanjivkumar Mhetre	1RV24EE027